

NexaMobiles — E-commerce Website Blueprint

1. Project Summary

Project name: NexaMobiles

Target market: Germany

Products: New mobile phones and accessories

Brands: Apple, Samsung, Vivo, Oppo, Realme

Language: English first

Goal: Build a fast MVP first, while keeping the structure strong enough for a fuller version later.

2. Core Customer Journey

Home



Shop / Brand / Category



Product Details



Add to Cart



Login / Signup



Checkout



Payment



Order Confirmation + Email

Customers can browse the full website without logging in.
Login is required only when they proceed to checkout.

3. MVP Features

Customer Features

- Responsive homepage
- Product listing page
- Brand-based browsing
- Category-based browsing
- Product details page
- Product variants
- Cart system
- Email/password signup and login
- Checkout flow
- Card payment
- PayPal payment
- Order confirmation page
- Order confirmation email
- Customer order history page

Admin Features

- Admin login
 - Dashboard
 - Add product
 - Edit product
 - Delete product
 - Manage stock
 - View orders
-

4. Product Data Model

Each product should support:

- Name
- Brand
- Category
- Price
- Description
- Main image
- Image gallery
- RAM
- Storage
- Color
- Stock
- Featured flag
- New arrival flag
- Variants

Example

```
iPhone 15
├─ 128GB / Black
├─ 128GB / Blue
└─ 256GB / Black
```

5. Recommended Tech Stack

Frontend

- React
- React Router
- Tailwind CSS
- Axios
- Context API or Zustand for cart/auth state

Backend

- Node.js
- Express.js
- JWT-based authentication
- bcrypt for password hashing

Database

- PostgreSQL

Database Toolkit

- Prisma ORM

Payments

- Stripe for card payments
- PayPal integration

Email

- Resend for transactional emails

Deployment Later

- Frontend: Vercel or Netlify
- Backend: Render / Railway / VPS
- Database: Managed PostgreSQL

6. High-Level Architecture

```
React Frontend
|
| REST API
v
Express Backend
|
| Prisma ORM
```

Why this architecture?

- Clear separation between frontend and backend
 - Easy to understand while learning
 - Good for future scaling
 - Lets the frontend and backend evolve independently
-

7. Main Pages

Public Pages

```
/
/shop
/brands/:brandSlug
/categories/:categorySlug
/products/:productSlug
/cart
/login
/signup
```

Protected Customer Pages

```
/checkout
/orders
/orders/:orderId
```

Admin Pages

```
/admin
/admin/products
/admin/products/new
/admin/products/:id/edit
```

/admin/orders
/admin/stock

8. Suggested Homepage Structure

1. Hero section
 - Premium headline
 - Featured phone
 - Shop now button
2. Brand strip
 - Apple, Samsung, Vivo, Oppo, Realme
3. Featured phones
4. New arrivals
5. Accessories section
6. Why choose us
 - Genuine products
 - Secure payment
 - Fast checkout
7. Footer

Design Direction

- Apple-style clean layout
 - Modern but not noisy
 - White / light background
 - Graphite text
 - Soft blue or electric accent color
 - Large product imagery
 - Generous spacing
 - Rounded cards
-

9. Database Tables

users

- id
- name
- email
- password_hash
- is_admin
- created_at
- updated_at

brands

- id
- name
- slug

categories

- id
- name
- slug

products

- id
- name
- slug
- description
- price
- brand_id
- category_id
- main_image
- is_featured
- is_new_arrival
- created_at
- updated_at

product_images

- id
- product_id
- image_url

product_variants

- id
- product_id
- ram
- storage
- color
- stock
- sku

cart_items

- id
- user_id
- product_variant_id
- quantity

orders

- id
- user_id
- total_amount
- status
- payment_status
- created_at

order_items

- id
- order_id
- product_id
- product_variant_id
- quantity
- unit_price

payments

- id
 - order_id
 - provider
 - provider_payment_id
 - amount
 - status
 - created_at
-

10. Suggested API Routes

Auth

```
POST /api/auth/register
POST /api/auth/login
GET /api/auth/me
```

Products

```
GET /api/products
GET /api/products/:slug
POST /api/products
PUT /api/products/:id
DELETE /api/products/:id
```

Brands and Categories

```
GET /api/brands
GET /api/categories
```

Cart

```
GET /api/cart
```

```
POST    /api/cart
PUT      /api/cart/:itemId
DELETE  /api/cart/:itemId
```

Orders

```
POST    /api/orders
GET      /api/orders/my-orders
GET      /api/orders/:id
GET      /api/admin/orders
```

Payments

```
POST    /api/payments/stripe/create-session
POST    /api/payments/paypal/create-order
POST    /api/payments/paypal/capture-order
```

11. Recommended Folder Structure

```
nexamobiles/
├─ client/
│  ├─ public/
│  └─ src/
│     ├─ assets/
│     ├─ components/
│     │  ├─ common/
│     │  ├─ layout/
│     │  └─ product/
│     │     └─ cart/
│     └─ pages/
│        ├─ public/
│        ├─ customer/
│        └─ admin/
└─ routes/
└─ services/
```

```
| | | | | store/
| | | | | hooks/
| | | | | utils/
| | | | | App.jsx
| | | | | main.jsx
| | | | | package.json
| | | | | tailwind.config.js
| | | | |
| | | | | server/
| | | | | | prisma/
| | | | | | | schema.prisma
| | | | | | | seed.js
| | | | | | src/
| | | | | | | config/
| | | | | | | controllers/
| | | | | | | middlewares/
| | | | | | | routes/
| | | | | | | services/
| | | | | | | utils/
| | | | | | | app.js
| | | | | | | server.js
| | | | | | package.json
| | | | | | .env
| | | | |
| | | | | docs/
| | | | | | NEXAMOBILES_BLUEPRINT.md
| | | | |
| | | | | README.md
| | | | | .gitignore
```

12. Frontend Folder Meaning

components/common

Reusable buttons, inputs, badges, loaders

components/layout

Navbar, footer, sidebar

components/product

ProductCard, ProductGallery, VariantSelector

components/cart

CartItem, CartSummary

- pages/public
 - Home, Shop, ProductDetails, Login, Signup
- pages/customer
 - Checkout, Orders
- pages/admin
 - Dashboard, ProductList, ProductForm, OrderList
- services
 - API calls to backend
- store
 - Cart/auth state management
- routes
 - React Router route setup

13. Backend Folder Meaning

- config
 - Database config, environment config
- controllers
 - Request handlers
- middlewares
 - Auth check, admin check, error handling
- routes
 - API route definitions
- services
 - Payment logic, email logic, business logic
- utils
 - Helpers
- prisma
 - Database schema and seed data

14. Step-by-Step Development Plan

Phase 1 — Setup

1. Create root project folder
2. Create `client` React app
3. Create `server` Express app
4. Configure Tailwind CSS
5. Configure PostgreSQL and Prisma
6. Add base environment variables

Phase 2 — Data Foundation

1. Create database schema
2. Add seed data for brands, categories, products, variants
3. Build backend product APIs
4. Test APIs

Phase 3 — Frontend Shopping Experience

1. Build main layout
2. Build homepage
3. Build shop page
4. Build product details page
5. Build cart system

Phase 4 — Authentication

1. Register API
2. Login API
3. Protected routes
4. Admin route protection

Phase 5 — Checkout and Orders

1. Checkout page
2. Create order flow
3. Stripe payment
4. PayPal payment
5. Order confirmation page
6. Confirmation email

Phase 6 — Admin Panel

1. Admin dashboard
2. Product CRUD
3. Stock management
4. Order management

Phase 7 — Polish

1. Responsive QA
2. Loading states
3. Empty states
4. Error handling
5. Form validation
6. Final UI cleanup

15. MVP Completion Checklist

Responsive homepage
Product catalog
Brand/category filtering
Product variants
Cart
Signup/login
Login required at checkout
Stripe payment
PayPal payment

Order confirmation page
Confirmation email
Admin dashboard
Product CRUD
Stock management
Order management

16. Later Enhancements

- Wishlist
- Reviews
- Coupons
- Product search
- Advanced filters
- Multi-language support
- Shipping module
- Returns module
- Sales analytics
- Staff roles